

University of Wisconsin
SCHOOL OF MEDICINE
AND PUBLIC HEALTH

Functional Data Analysis

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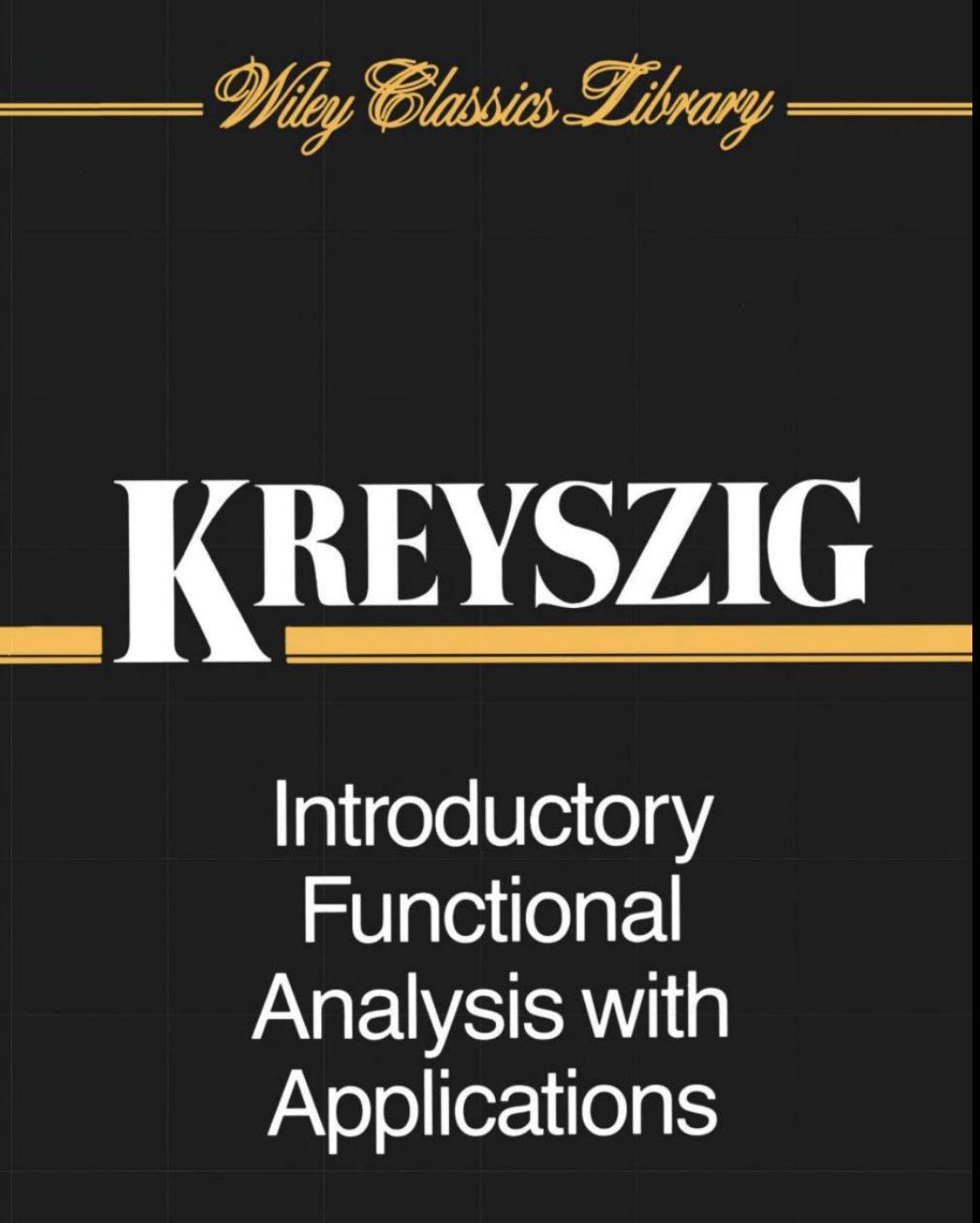
Unit goals

Able to model functional data using basis functions

Able to fit the basis functions to data using LSE

Functional Data Analysis on [0,1]

Question: Why we are limiting in [0,1]?

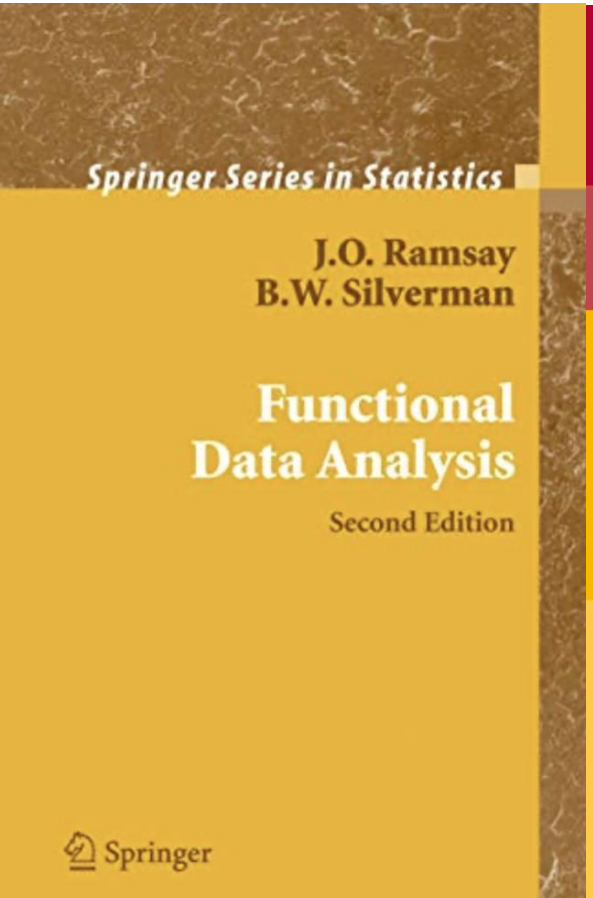


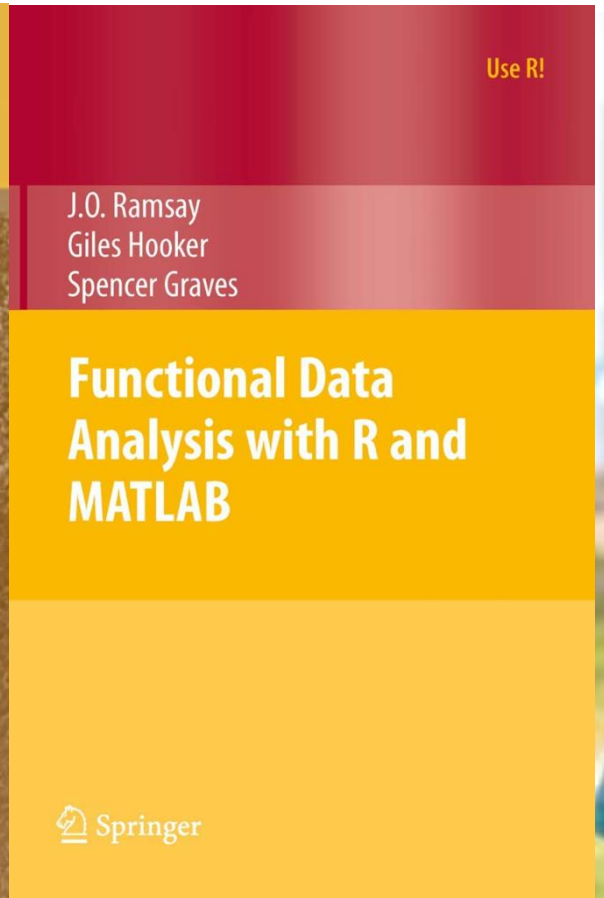
Functional analysis is the area of mathematics that studies spaces of functions and the linear operators acting on them.

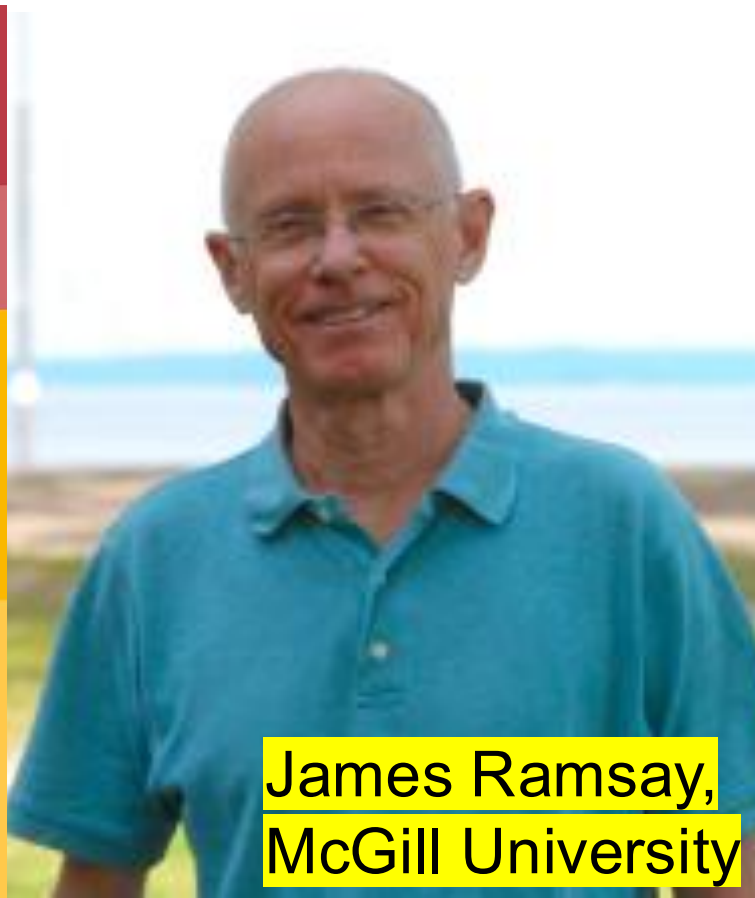
History [edit]

Wikipedia

Functional data analysis has roots going back to work by Grenander and Karhunen in the 1940s and 1950s.[1][2][3][4] They considered the decomposition of square-integrable continuous time stochastic process into eigencomponents, now known as the Karhunen-Loève decomposition. A rigorous analysis of functional principal components analysis was done in the 1970s by Kleffe, Dauxois and Pousse including results about the asymptotic distribution of the eigenvalues.[5][6] More recently in the 1990s and 2000s the field has focused more on applications and understanding the effects of dense and sparse observations schemes. The term "Functional Data Analysis" was coined by James O. Ramsay.[7]







Deformable Pattern Theory

Anatomical variability is modeled by deforming a common template image,

$I_i = I_{\text{temp}} \circ \phi_i^{-1},$

where I_{temp} is the template and ϕ_i is a smooth deformation map. Statistical analysis is then performed on the deformation fields ϕ_i, which represent subject-specific anatomical variation.



Michel I. Miller

We will fit a functions in [0,1]

step 1: Set up Hilbert space
step 2: Model setup using basis
step 3: Perform LSE

The **Hilbert space** named after David Hilbert, generalizes the notion of Euclidean space.
A Hilbert space is an abstract vector space possessing the structure of an inner product that allows length and angle to be measured.



Vector Spaces

A vector space M has the following properties.



- You can add elements.
- You can multiply elements by real numbers.
- Associative and distributive properties:

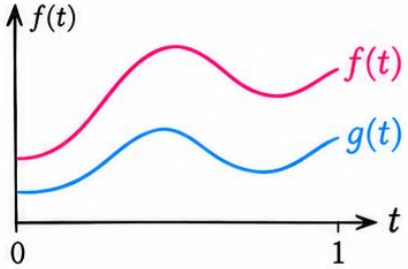
$$u + (v+w) = (u+v) + w$$
$$a(u+v) = au + av$$

We can add and scale vectors, and they satisfy the usual rules.

Examples of vector spaces

$L^2([0,1])$

The space of square integrable functions



$$L^2([0,1]) = \left\{ f : [0,1] \to \mathbb{R} \;\middle|\; \int_0^1 |f(t)|^2 dt < \infty \right\}$$

Exercise:

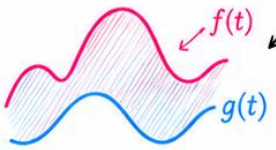
Check if $L([0,1])$ is a vector space.

Functions with finite energy form a vector space.

Inner product

Inner product associates each pair of elements in the space with a number.

$$\langle f, g \rangle = \int_0^1 f(t)g(t)dt$$



Measures similarity between functions.

$$L_2\text{-norm} \quad \|f\| = \sqrt{\int_0^1 f(t)^2 dt}$$
$$= \sqrt{\langle f, f \rangle}$$

Inner product space

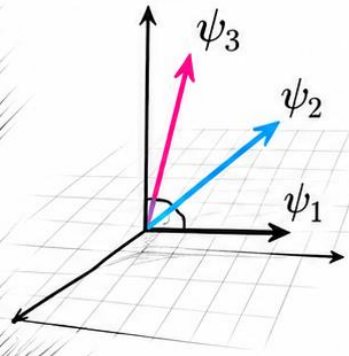
= vector space + inner product

The inner product gives a notion of length and angle between functions.

Exercise:

Check if $L([0,1])$ is an inner product space.

Orthonormal Basis in a Vector Space



Orthonormal basis

$$\langle \psi_i, \psi_j \rangle = \delta_{ij}$$

Basis vectors are mutually orthogonal and have unit norm.

$$\|\psi_j\| = 1$$
$$\langle \psi_i, \psi_j \rangle = 0 \ (i \neq j)$$

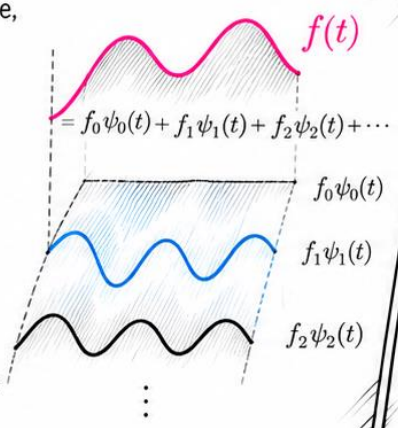
Basis expansion

For any f in the space,

$$f_j = \langle f, \psi_j \rangle$$

$$f = \sum_j f_j \psi_j$$

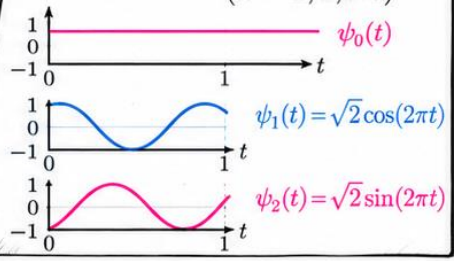
The coefficients f_j describe how much f aligns with each basis function ψ_j.



Example: $L^2([0,1])$

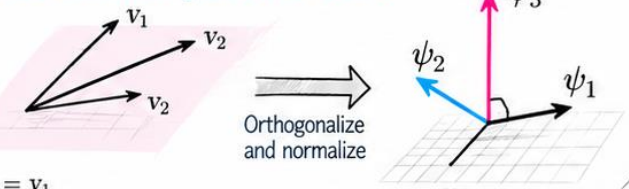
A classic orthonormal basis is the Fourier basis on $[0, 1]$.

$$\psi_0(t) = 1$$
$$\psi_{2n-1}(t) = \sqrt{2}\cos(2\pi n t)$$
$$\psi_{2n}(t) = \sqrt{2}\sin(2\pi n t)$$
$$(n = 1, 2, \ldots)$$



How to obtain an orthonormal basis

Gram–Schmidt process (idea)



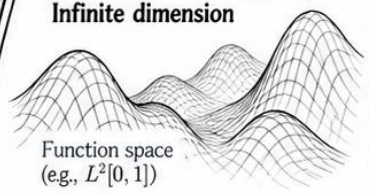
$$u_1 = v_1$$
$$u_2 = v_2 - \mathrm{proj}_{u_1}(v_2)$$
$$u_3 = v_3 - \mathrm{proj}_{u_1}(v_3) - \mathrm{proj}_{u_2}(v_3)$$
$$\vdots$$
$$u_k = v_k - \sum_{j=1}^{k-1} \mathrm{proj}_{u_j}(v_k)$$

$$\psi_k = \frac{u_k}{\|u_k\|}$$

This produces an orthonormal set $\{\psi_k\}$ spanning the same space as $\{v_k\}$.

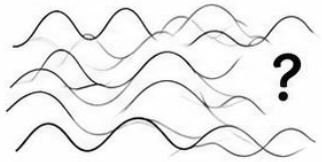
Why Gram–Schmidt is not used in high-dimensional function spaces

Infinite dimension



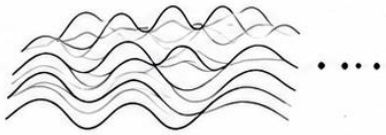
Gram–Schmidt is designed for finite-dimensional settings. It is not practical for infinitely many functions.

No canonical starting set



The result depends on the choice and order of the initial functions, so it is not unique.

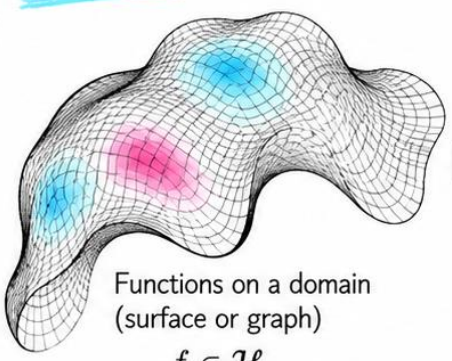
Computational and interpretability issues



In high-dimensional spaces, the resulting basis functions are often complicated and lack useful structure, making them less interpretable and less convenient for analysis.

In practice, we use structured orthonormal bases (e.g., Fourier, wavelets) derived from the properties of the space, rather than a general Gram–Schmidt construction.

Basis from spectral geometry



Functions on a domain
(surface or graph)

$f \in \mathcal{H}$

Self-adjoint operator

$$\langle Lf, g \rangle = \langle f, Lg \rangle$$

In the matrix form

$$g^\top L f = f^\top L g$$

Geometric operators (examples)

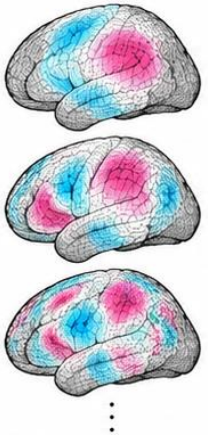
$-\Delta$ (Laplace–Beltrami on a surface)

L (graph Laplacian)

Δ (Hodge Laplacian)

$\vdots$

Eigenfunctions of L



ψ_1 (smooth)

ψ_2

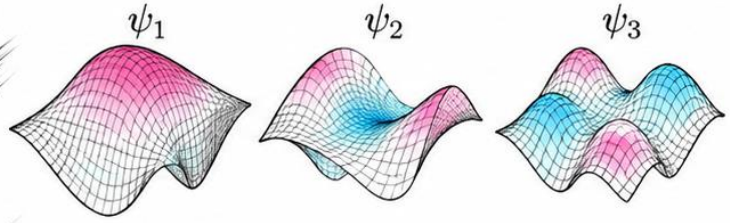
ψ_3

$\vdots$

Exercise: 1) Determine matrix L satisfying this condition.

Basis from spectral geometry

For a self-adjoint operator $L\psi_j = \lambda_j \psi_j$



Orthogonality of eigenfunctions

For $\lambda_i \neq \lambda_j$,

$$\langle L\psi_i, \psi_j \rangle = \langle \psi_i, L\psi_j \rangle$$
$$\lambda_i \langle \psi_i, \psi_j \rangle = \lambda_j \langle \psi_i, \psi_j \rangle$$
$$\Rightarrow \quad \langle \psi_i, \psi_j \rangle = 0$$

Unit normalization

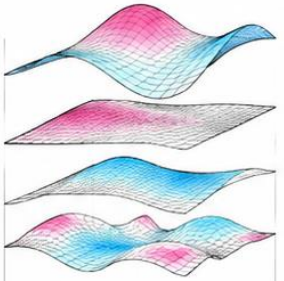
$$\langle \psi_j, \psi_j \rangle = 1$$

$\{\psi_j\}$ form an ***orthonormal basis.***

Exercise: Prove they are orthonormal basis.

Basis expansion

$$f = \sum_j \langle f, \psi_j \rangle \psi_j$$



f

$\langle f, \psi_1 \rangle \psi_1$

$\langle f, \psi_2 \rangle \psi_2$

$\langle f, \psi_3 \rangle \psi_3$

$\vdots$

Cosine basis out of 1D Laplacian

1D Laplacian

$$\Delta = \frac{\partial^2}{\partial t^2}$$

with Neumann boundary condition

$$f'(0) = f'(1) = 0$$



Eigenvalue problem

$$\Delta\psi_l(t) = \frac{\partial^2}{\partial t^2}\psi_l(t) = \lambda_l\psi_l(t)$$

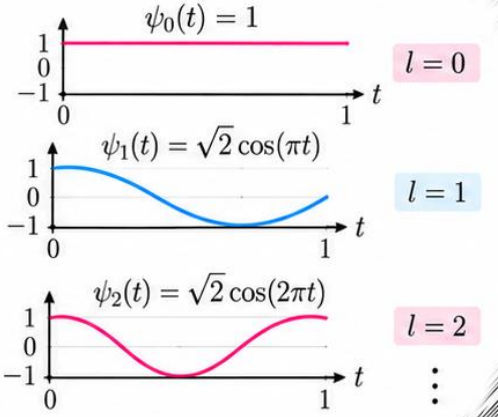
Solutions (Neumann):

$$\psi_0(t) = 1, \quad \psi_l(t) = \sqrt{2}\cos(l\pi t) \quad (l = 1, 2, \ldots)$$

Eigenvalues:

$$\lambda_0 = 0, \quad \lambda_l = -(l\pi)^2 \quad (l \geq 1)$$

Eigenfunctions



Self-adjoint operator

$$\langle Lf, g\rangle = \langle f, Lg\rangle$$

For $L = \frac{\partial^2}{\partial t^2}$ with $f'(0) = f'(1) = 0$,

$$\langle Lf, g\rangle = \int_0^1 f''(t)g(t)\,dt$$

$$= [f'(t)g(t)]_0^1 - \int_0^1 f'(t)g'(t)\,dt$$

$$= \int_0^1 f(t)g''(t)\,dt = \langle f, Lg\rangle$$

Thus L is a self-adjoint operator under Neumann boundary conditions.

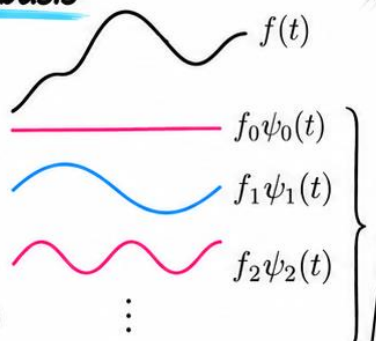
Expansion in the cosine basis

$$f(t) = \sum_{l=0}^{\infty} f_l\psi_l(t)$$

Coefficients (inner product)

$$f_l = \langle f, \psi_l\rangle = \int_0^1 f(t)\psi_l(t)\,dt$$

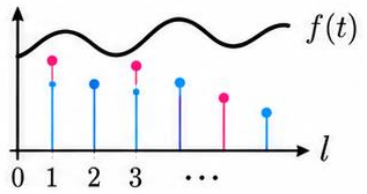
$$= \begin{cases} \int_0^1 f(t)\,dt & (l = 0) \\ \sqrt{2}\int_0^1 f(t)\cos(l\pi t)\,dt & (l \geq 1) \end{cases}$$



The coefficients f_l can be computed in the least squares estimation.

Questions

1. What is the difference between our method and Fourier transform widely used in signal processing?



2. How to compute the integral using LSE?

Exercise: Check if it is a self-adjoint operator

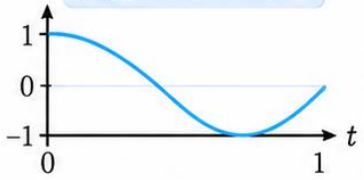
Is orthogonal basis unique?

$[0,1]$ $\quad \psi_0(t) = 1, \quad \psi_l(t) = \sqrt{2}\cos(l\pi t) \quad (l \geq 1)$

$\psi_0(t) = 1$



$\psi_1(t) = \sqrt{2}\cos(\pi t)$



$$\langle \psi_i, \psi_j \rangle = \int_0^1 \psi_i(t)\psi_j(t)\,dt = \delta_{ij}$$

Answer: No, it is not unique.

If $\{\psi_i\}$ is an orthonormal basis, then so is

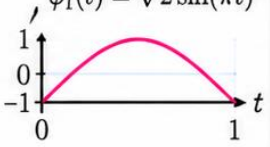
$$\phi_i = \sum_j U_{ji}\psi_j$$

where U is an orthogonal matrix ($U^{\mathrm{T}}U = I$).

Example:
new orthogonal basis

$\phi_0(t) = 1,$
$\phi_1(t) = \sqrt{2}\sin(\pi t)$

$\phi_1(t) = \sqrt{2}\sin(\pi t)$



Many orthogonal bases in [0,1]

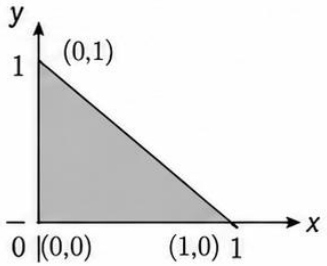
- Cosine basis (Neumann)
  $\psi_0 = 1, \quad \psi_l = \sqrt{2}\cos(l\pi t)$
- Sine basis (Dirichlet)
  $\chi_l(t) = \sqrt{2}\sin(l\pi t) \quad (l \geq 1)$
- Linear combinations (rotations)
  $\phi_l(t) = \cos\theta\,\psi_l(t) + \sin\theta\,\chi_l(t)$

Thus, an orthogonal basis is not unique.
There are infinitely many different bases in [0,1].

How to find basis for an arbitrary manifold

Example: a triangular domain

Domain Ω (triangle)



Find the self-adjoint operator

Consider the Laplacian on Ω:

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

with appropriate boundary condition

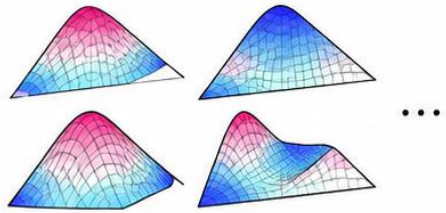
- Dirichlet: $u = 0 \quad$ on $\partial\Omega$
- Neumann: $\dfrac{\partial u}{\partial n} = 0 \quad$ on $\partial\Omega$

Solve the eigenvalue problem

$$-\Delta\phi = \lambda\phi \quad \text{in } \Omega$$
$$+ \text{ boundary condition}$$

Then $\{\phi_i\}$ gives an orthogonal basis of $L^2(\Omega)$.

Eigenfunctions on the triangle



If no analytical solution

- Use numerical methods
  - finite element method (FEM)
  - finite difference method
  - spectral method
- Compute eigenvalues and eigenfunctions numerically.

Project (suggestion)

If you cannot find a mathematical solution, approximate it numerically. This is a potential class project.

Project: This is a potential class project and connect to Dirichlet distribution.

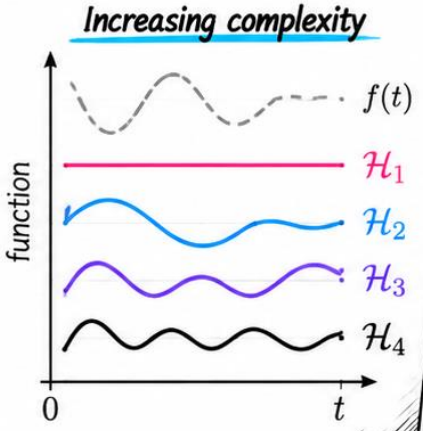
Filtration: nested functional subspace

Filtration: nested functional subspace

$$\mathcal{H}_k = \left\{ \sum_{l=0}^{k} c_l \psi_l(t) : c_l \in \mathbb{R} \right\}$$

Nested subspaces

$$\mathcal{H}_1 \subset \mathcal{H}_2 \subset \cdots$$



Example: cosine basis on [0,1]

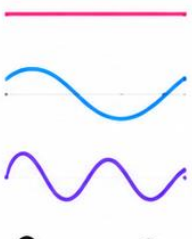
$$\psi_0(t) = 1, \quad \psi_l(t) = \sqrt{2}\cos(l\pi t) \quad (l \geq 1)$$

$\mathcal{H}_1 = \mathrm{span}\{\psi_0\}$ $\quad f(t) = c_0\psi_0(t)$

$\mathcal{H}_2 = \mathrm{span}\{\psi_0, \psi_1\}$ $\quad f(t) = c_0\psi_0(t) + c_1\psi_1(t)$

$\mathcal{H}_3 = \mathrm{span}\{\psi_0, \psi_1, \psi_2\}$ $\quad f(t) = \sum_{l=0}^{2} c_l\psi_l(t)$

$\mathcal{H}_4 = \mathrm{span}\{\psi_0, \psi_1, \psi_2, \psi_3\}$ $\quad f(t) = \sum_{l=0}^{3} c_l\psi_l(t)$



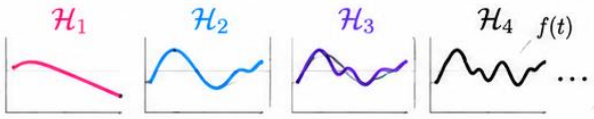
⋮ ⋮

Why build a filtration?

We approximate a function space by a sequence of finite-dimensional subspaces:

$$\mathcal{H}_1 \subset \mathcal{H}_2 \subset \cdots$$

This gives a multi-scale representation of functions. As k increases, $\mathcal{H}_k$ captures more detailed features.



Application to persistent homology

From data (e.g., a function, a shape, or a point cloud), we build a filtration of spaces:

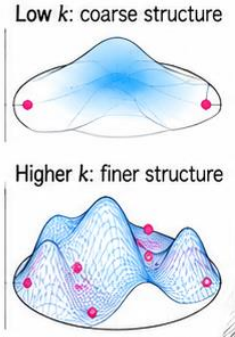
$$\mathcal{H}_1 \subset \mathcal{H}_2 \subset \cdots$$

↓

Compute homology at each scale k

↓

Persistent homology reveals features that persist across scales.



Project

We can build persistent homology out of this space.

Idea:

- Choose an appropriate basis $\{\psi_l\}$.
- Construct the nested subspaces $\{\mathcal{H}_k\}$.
- Analyze the topological features using persistent homology.

Project: We can build persistent homology out of function space.

Statistical model on functional data

$$\zeta(t) = \mu(t) + \epsilon(t)$$

random field

$$\mu \in \mathcal{L}^2[0,1]$$

$$\mu = \sum_{l=0}^{k} c_l \psi_l(t), \quad c_l\text{: random variables}$$

Function estimation

$$\widehat{\mu} = \arg\min_{f \in \mathcal{H}_k} \left\| f - \zeta(t) \right\|^2 \qquad f(t) = \sum_{l=0}^{k} c_l \psi_l(t)$$

$$J(c_0, \ldots, c_k) = \left\| \sum_{l=0}^{k} c_l \psi_l - \xi \right\|^2$$

$$= \left\langle \sum_{l=0}^{k} c_l \psi_l - \xi \,,\, \sum_{m=0}^{k} c_m \psi_m - \xi \right\rangle$$

$$= \sum_{l,m=0}^{k} c_l c_m \langle \psi_l, \psi_m \rangle - \sum_{l=0}^{k} c_l \langle \psi_l, \xi \rangle - \sum_{m=0}^{k} c_m \langle \psi_m, \xi \rangle$$

$$= \sum_{l=0}^{k} c_l^2 - \sum_{l=0}^{k} c_l \xi_l - \sum_{l=0}^{k} c_l \xi_l$$

Function estimation

$$\widehat{\mu} = \arg \min_{f \in \mathcal{H}_k} \left\| f - \zeta(t) \right\|^2$$

$$I(c_0, \ldots, c_k) = \sum_{l=0}^{k} c_l^2 - 2 \sum_{l=0}^{k} c_l \xi_l$$

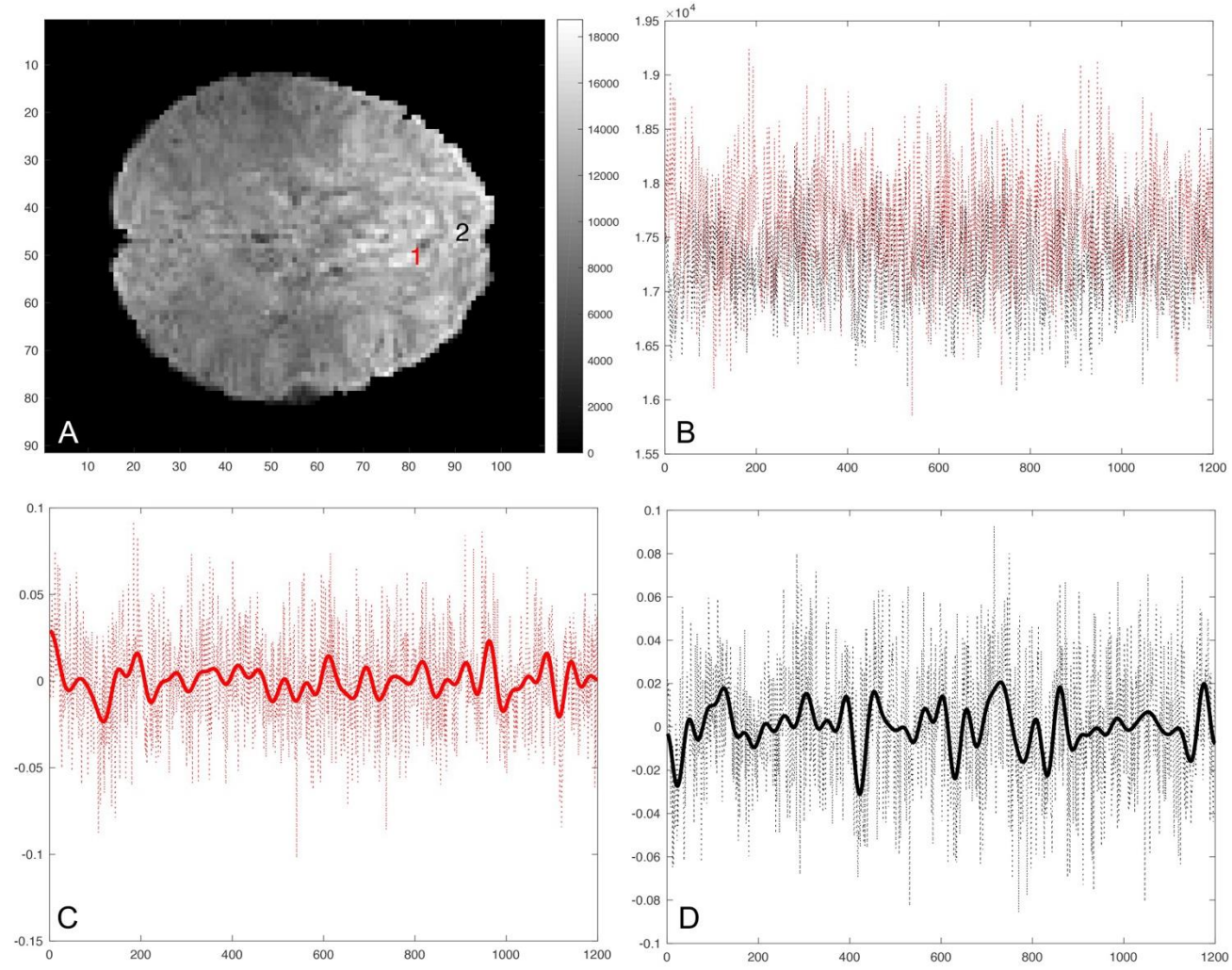
$$\frac{\partial I}{\partial c_j} = 2 c_j - 2 \xi_j = 0$$

$$c_j = \xi_j = \langle \xi, \psi_j \rangle$$

Fourier transform:

$$c_l = \langle \zeta, \psi_l \rangle$$ Final outcome

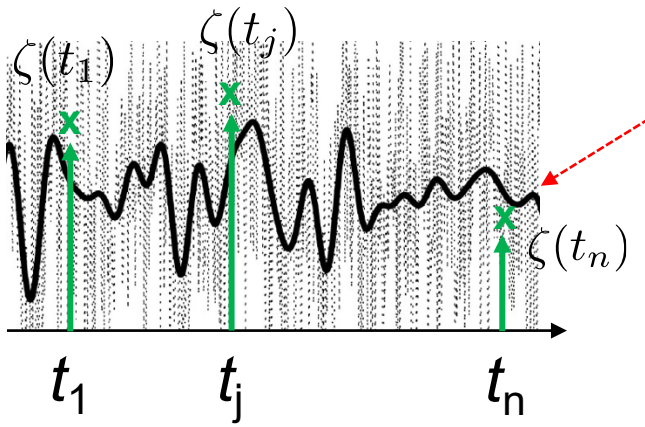
Cosine series representation



59-degree expansion

The actual application of the representation in brain imaging data:

Least squares estimation



$$\mu(t) = \sum_{l=0}^{k} c_l \psi_l(t)$$

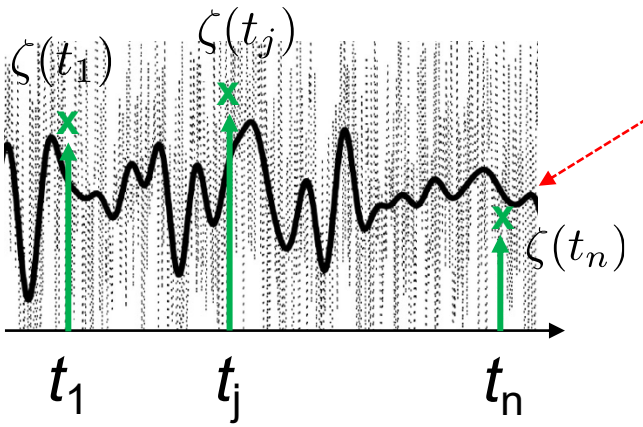
Design matrix of basis functions.

$$\Psi_{n\times(k+1)} = \begin{pmatrix} \psi_0(t_1) & \psi_1(t_1) & \cdots & \psi_k(t_1) \\ \psi_0(t_2) & \psi_1(t_2) & \cdots & \psi_k(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_0(t_n) & \psi_1(t_n) & \cdots & \psi_k(t_n) \end{pmatrix}$$

Exercise: Check if $\Psi^\top \Psi = I_k$

You will not get orthogonality. Why?

Least squares estimation



$$\mu(t) = \sum_{l=0}^{k} c_l \psi_l(t)$$

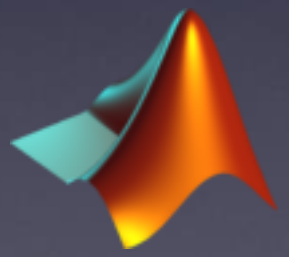
$$Y = \Psi C$$

Fourier transforms
(Fourier coefficients)

$$C = (\Psi^\top \Psi)^{-1} \Psi^\top Y$$

Question: 1) Can we do L1-norm based estimation here?
2) Is FFT faster than LSE?
3) Is FFT more accurate than LSE?

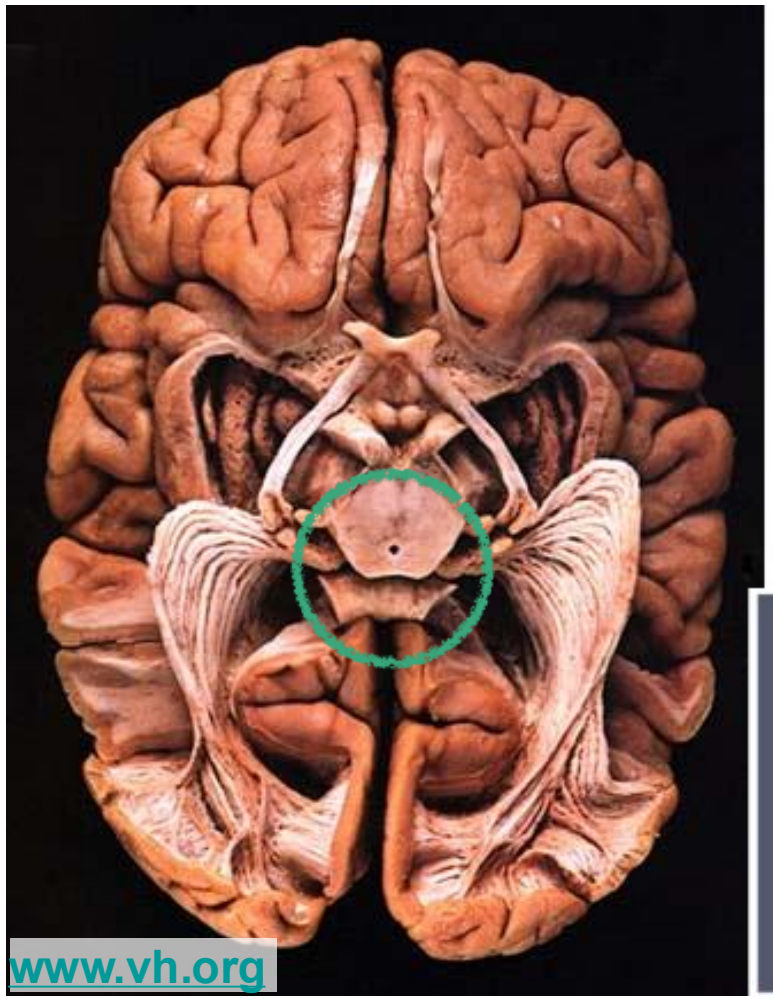
Application to 3D curve modeling

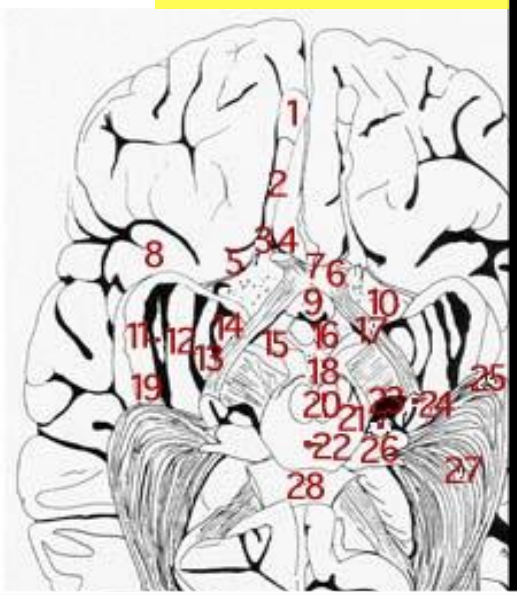


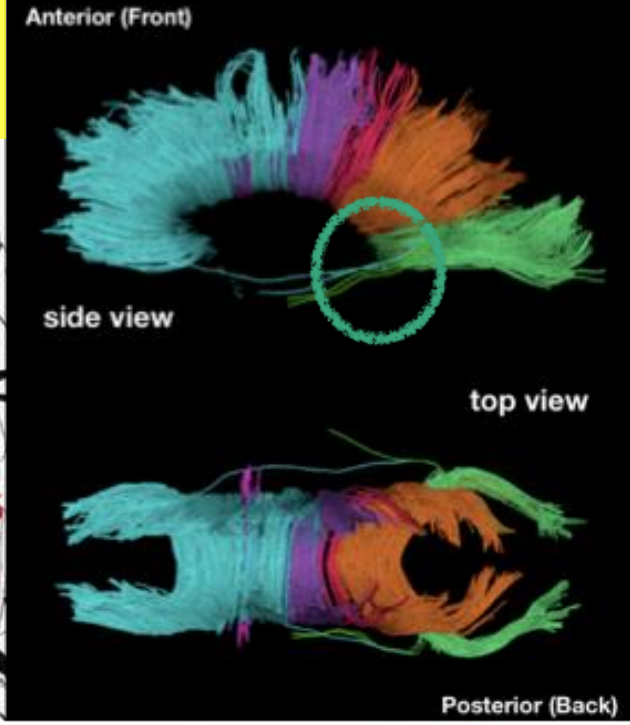
Matlab demo
https://github.com/laplcebeltrami/BMI768/tree/main/FDA/matlab

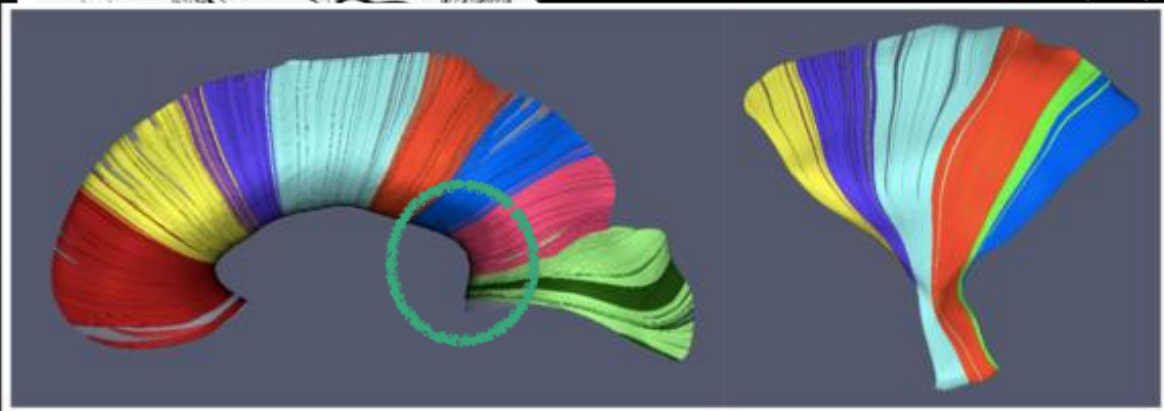
White matter fibers

James Gee
Univ. of Penn.









Fibers passing through the splenium of the corpus callosum

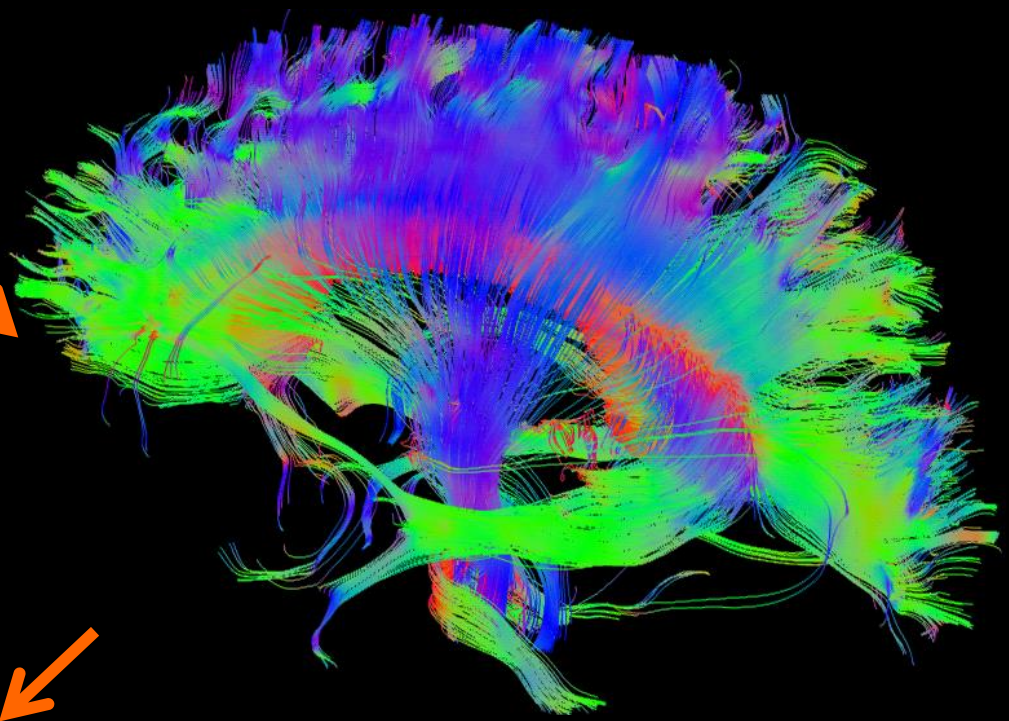




White matter fiber tractography

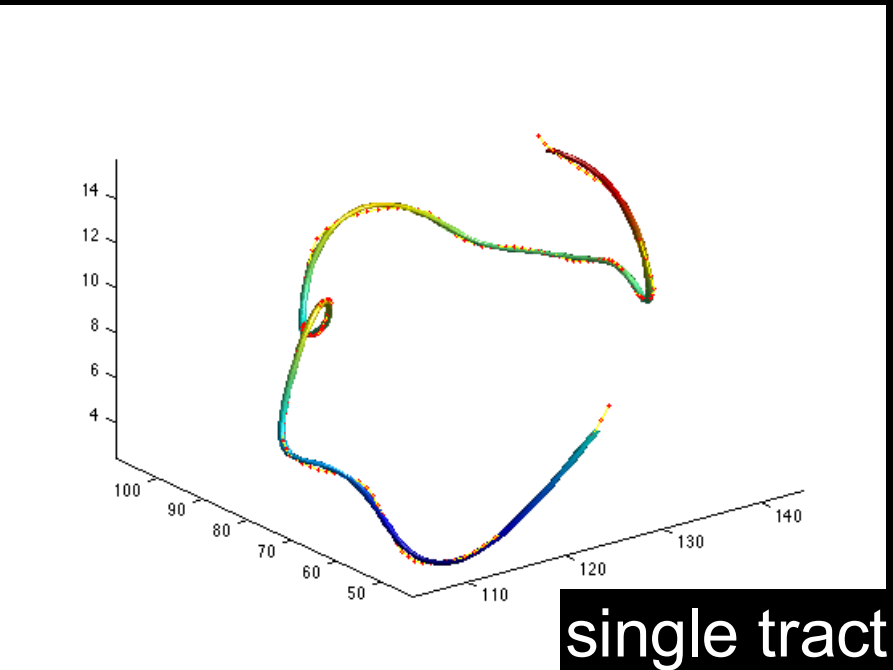


Real brain



Tractography result

0.01-10 million tracts



single tract → parameterization

Parametric model of white fiber tracts

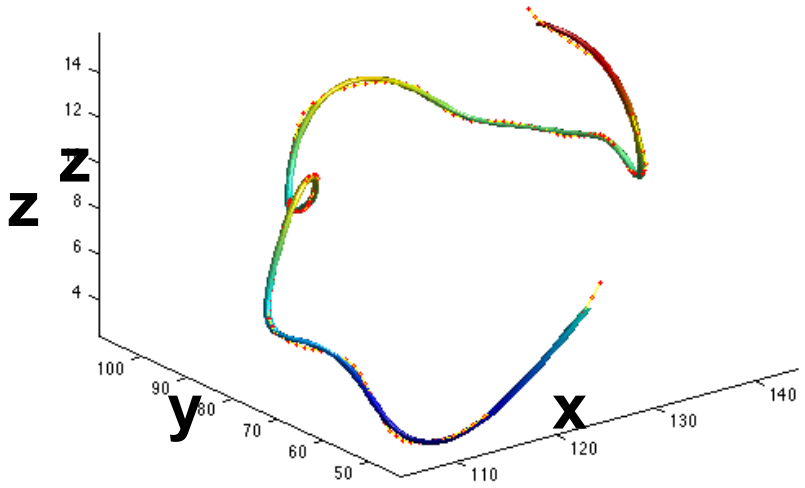
Clayden et al. IEEE TMI 2007
Cubic B-spline is used to model and match tracts.

Batchelor et al. MRM 2006
Sine and cosine Fourier descriptors are used to extract global shape features for classification

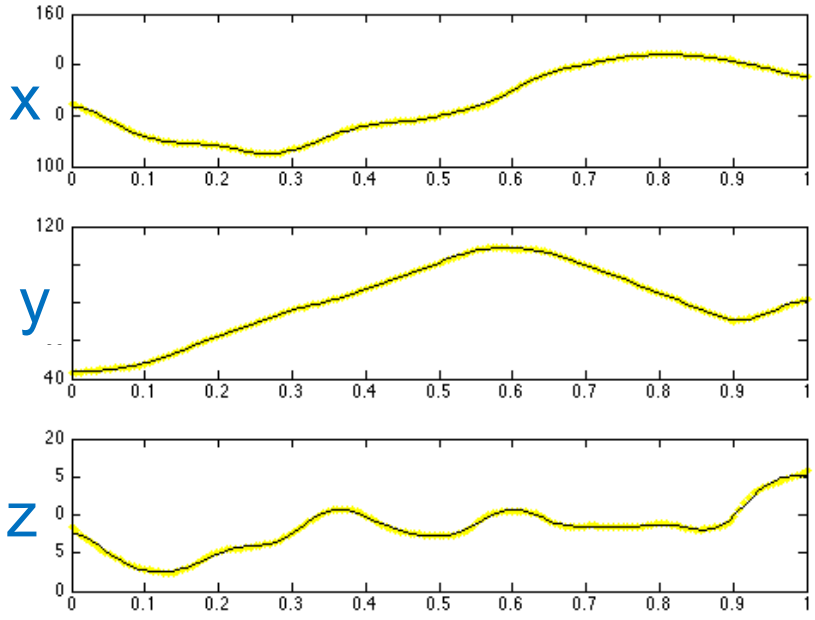
Chung, M.K., Adluru, N., Lee, J.E., Lazar, M., Lainhart, J.E., Alexander, A.L., 2010. Cosine series representation of 3D curves and its application to white matter fiber bundles in diffusion tensor imaging. Statistics and Its Interface. 3:69-80

Project*:* Build local basis like wavelets from global basis? → **Graph Slepian**

Regression on 3D curves



parameterization



x	y	z
88.1799	56.6336	5.7367
-12.4775	-11.2552	-2.0791
2.4336	-15.4428	-0.4021
4.3956	2.2733	-0.9354
-0.0106	-0.0674	0.6999
2.1773	-2.4194	-0.1176
0.5808	0.8390	1.2942
0.0615	-0.1893	0.1188
-0.2629	0.7524	0.1089
0.7909	-0.7276	-0.1901
0.5458	0.6236	0.6939
0.4295	-0.4337	0.2185
0.2150	0.4157	0.0254
0.1584	-0.1973	0.0762
-0.1557	0.2466	-0.1086
0.0632	-0.0978	-0.0208
0.0389	-0.0143	-0.0284
-0.0014	-0.1193	0.1970
0.0004	0.0129	-0.0198
0.1342	0.0002	0.0260

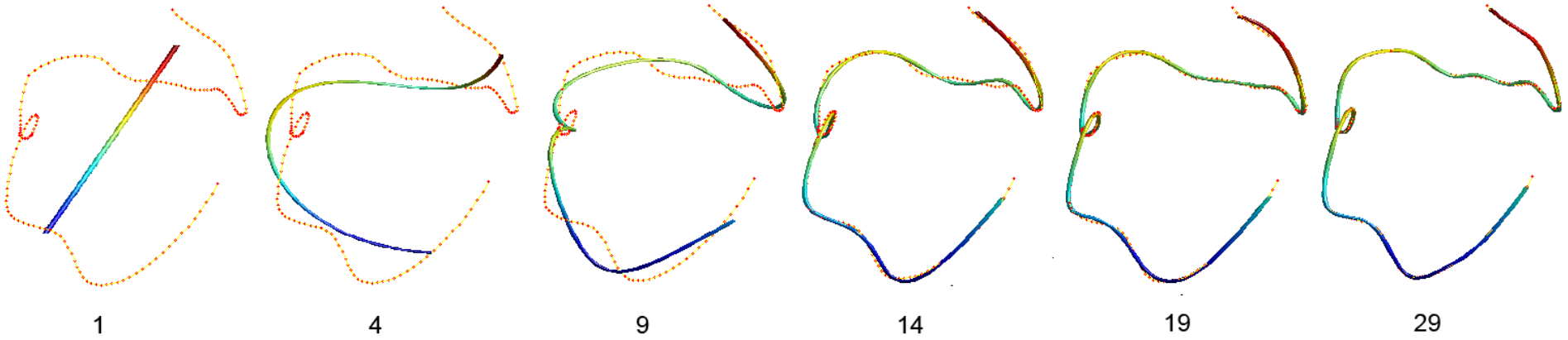
Any tract can be compactly parameterized with only 60 coefficients.

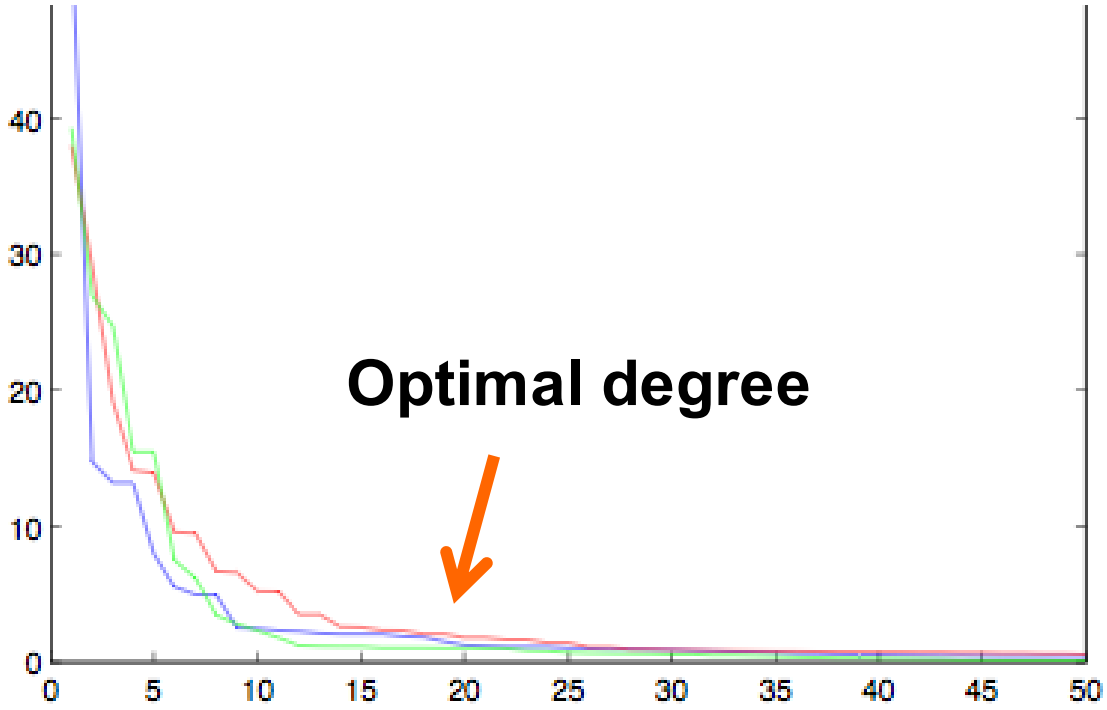
Curve registration is done by matching these parameters.

basis expansion

$$(x, y, z)' = \sum_{l=0}^{19} \beta_l \cos(l\pi t)$$

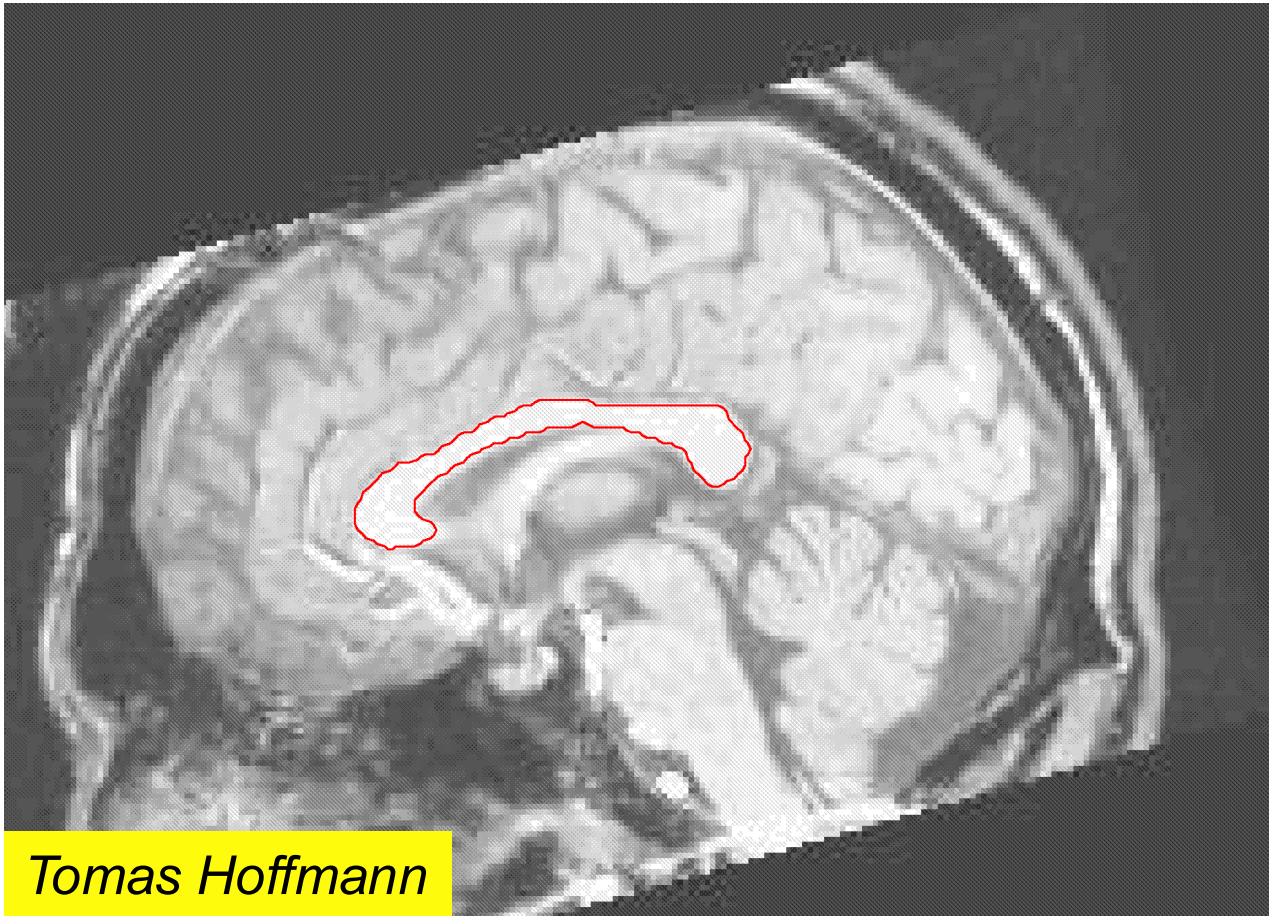
Cosine series representation





Optimal degree chosen using the forward model selection method.

Question: corpus callosum boundary - 1D closed curve



(x,y,z)-coordinates can be mapped to a circle and then parameterized. But how?

Self exercise

1) Following the lecture and studying the MATLAB code, write a MATLAB function that fits a function in interval [0,2].

2) Instead of cosine basis, use Hermite polynomial basis (hermiteH.m) and fit white matter fiber tracts.

Project: Which basis is better?

Advanced problems

Develop a method to fit a functional data on a unit circle (circular data) → project

Question: Can we use cosine basis? Can we use Hermite polynomial for fitting circular functional data?

Suggestion: The spectral geometry method can do closed-curve parameterization.

Unit outcomes

Can model functional data using basis functions

Can fit the basis functions to data using LSE

Can use the eigenfunction of Laplacian in fitting functional data